

Signal Diode-Based Fire Alarm

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A simple signal diode can be used to build a highly sensitive fire alarm. Silicon diodes like OA71 and 1N34 respond to infrared radiation and heat from fire by generating reverse current across their terminals. In reverse-bias mode, this effect is more significant. Typically, for each degree rise in temperature, the diode generates 2 mV. This characteristic is exploited in this circuit to sense fire. The circuit can detect fire from a distance of up to around 30 cm.

Circuit and working

Fig. 1 shows the circuit of signal diode based fire sensor. The circuit is powered by a standard transformer power supply using a step-down transformer X1 (230V AC primary to 9V-0-9V, 500mA secondary), rectifier diodes D1 and D2, and smoothing capacitor C1. A 9V rechargeable battery is added for power backup in the event of power

failure resulting from fire breakout. When mains power is available, the battery charges through diode D3 and resistor R1. When mains power fails, D3 reverse biases and D4 forward biases to take up the load.

Signal diode OA71 (D5) is used as a heat and light sensor. It readily accepts the infrared light from the fire and generates voltage across it. This voltage is fed to op-amp CA3140 (IC1), which is used as a sensitive inverting amplifier.

CA3140 (IC1) is a BiMOS op-amp IC that can operate down to zero volt output. The highest output available from IC1 is 2.25 volts less than the supply voltage at pin 7. IC1 has gate-protected MOSFET inputs with a



very high input impedance. An input current as low as 5 pico-amperes can change its output state.

PARTS LIST

Semiconductors:

IC1	- CA3140 op-amp
T1	- BC547 npn transistor
LED1	- 5mm LED
D1-D4	- 1N4007 rectifier diode
D5	- OA71 signal diode
ZD1	- 3.1V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 68-ohm
R2, R3, R5-R7	- 10-mega-ohm
R4	- 3.9-mega-ohm
R8	- 10-kilo-ohm
R9	- 22-kilo-ohm
R10, R11	- 100-ohm
VR1	- 4.7-kilo-ohm preset

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2	- 22 μ F, 25V electrolytic
C3	- 10pF ceramic disk
C4	- 10 μ F, 25V electrolytic

Miscellaneous:

X1	- 230C AC primary to 9V-0-9V, 500mA Secondary transformer
PZ1	- Piezo buzzer
	- 9V battery (chargeable)

Test Points

Test point	Details
TP0	GND
TP1	9V
TP2	6V (approx.)

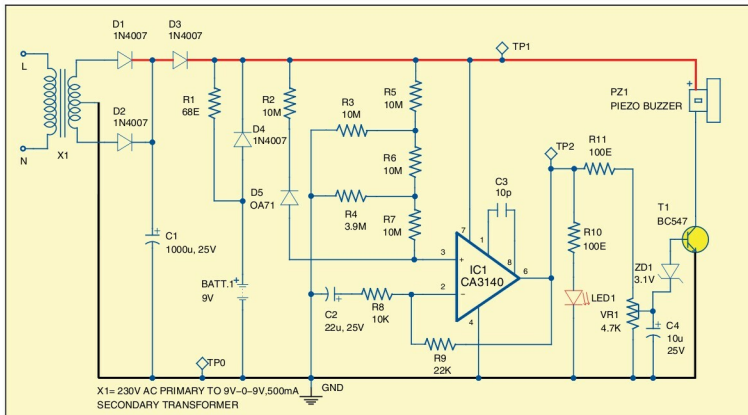


Fig. 1: Circuit of signal diode based fire sensor